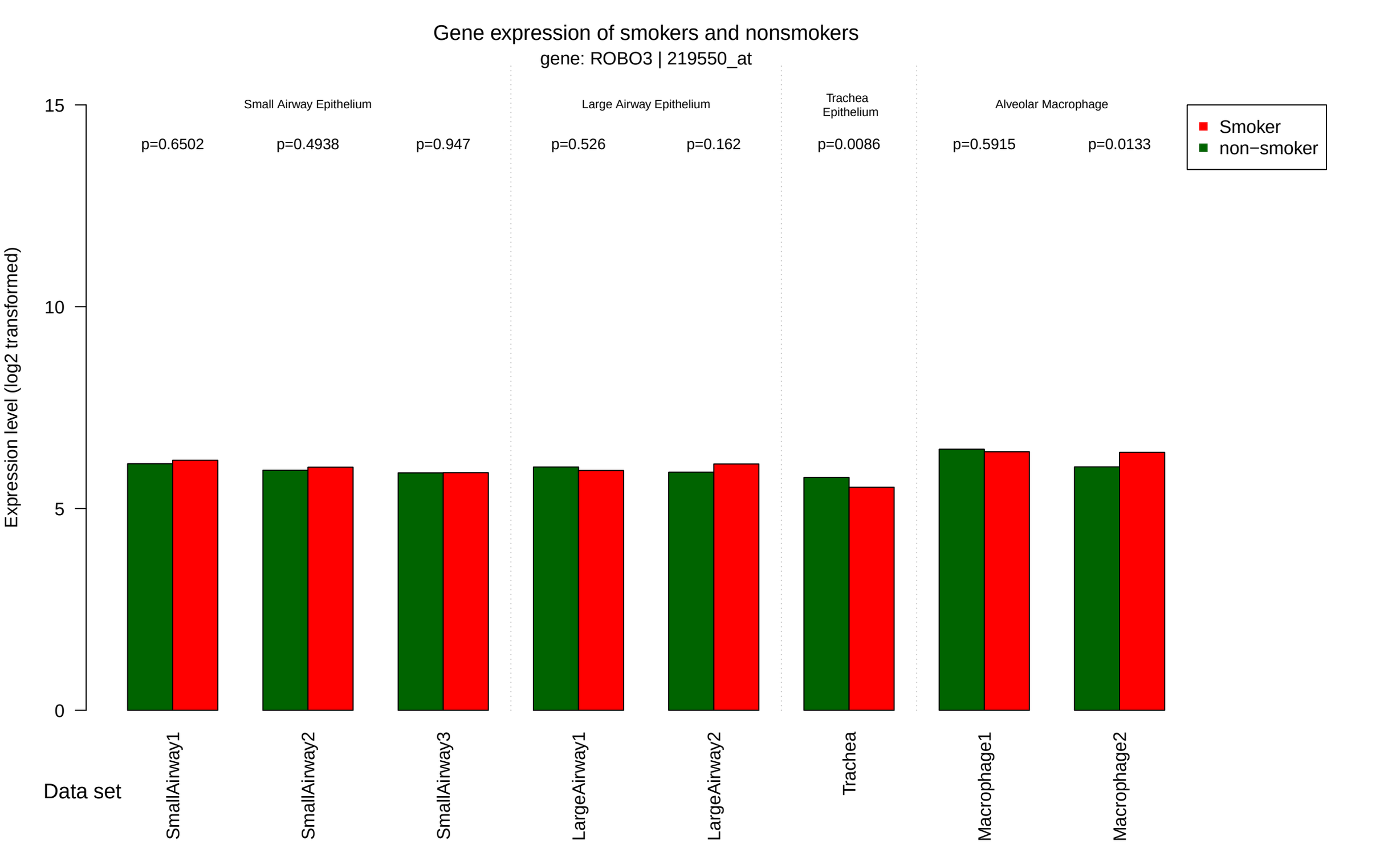
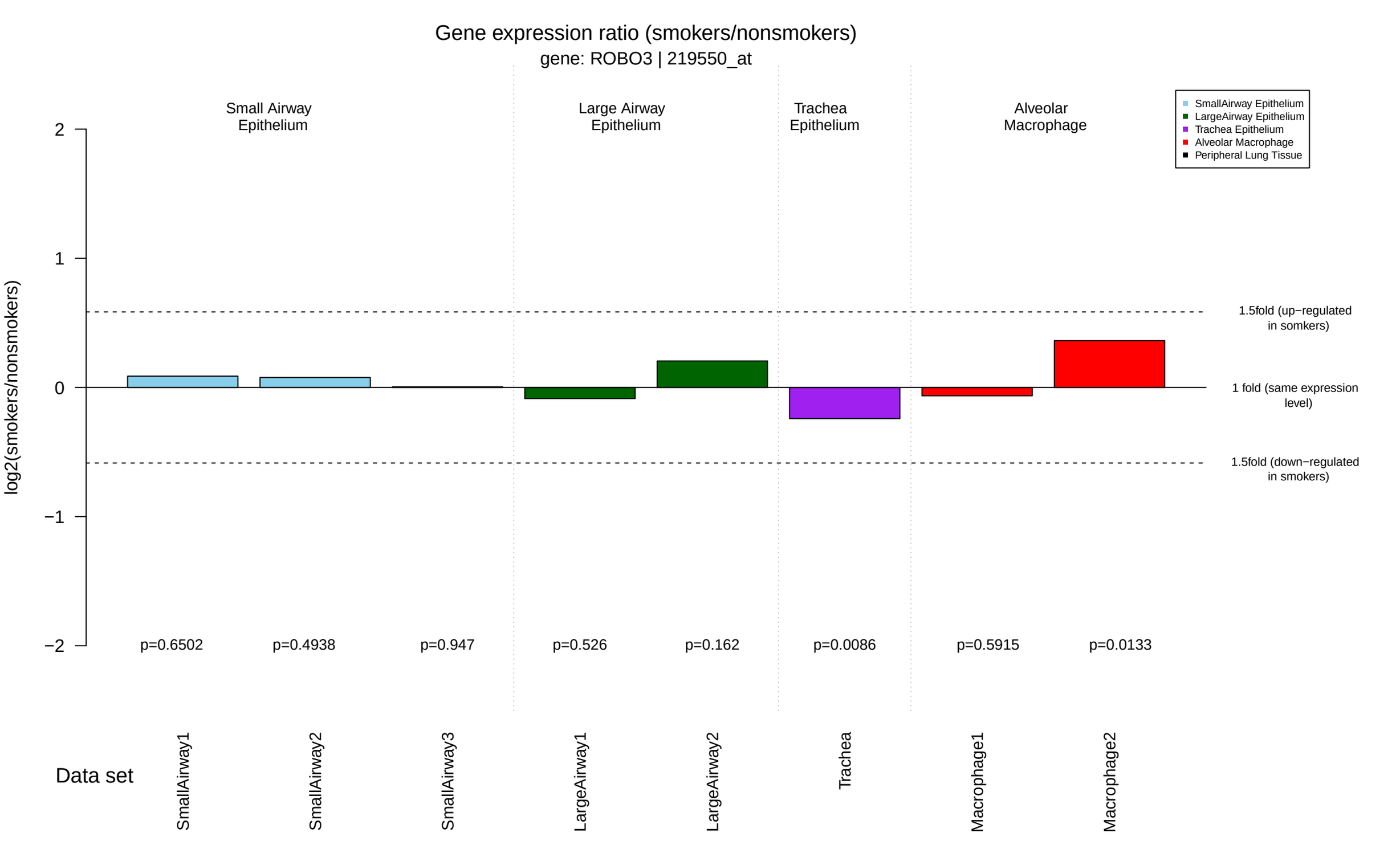
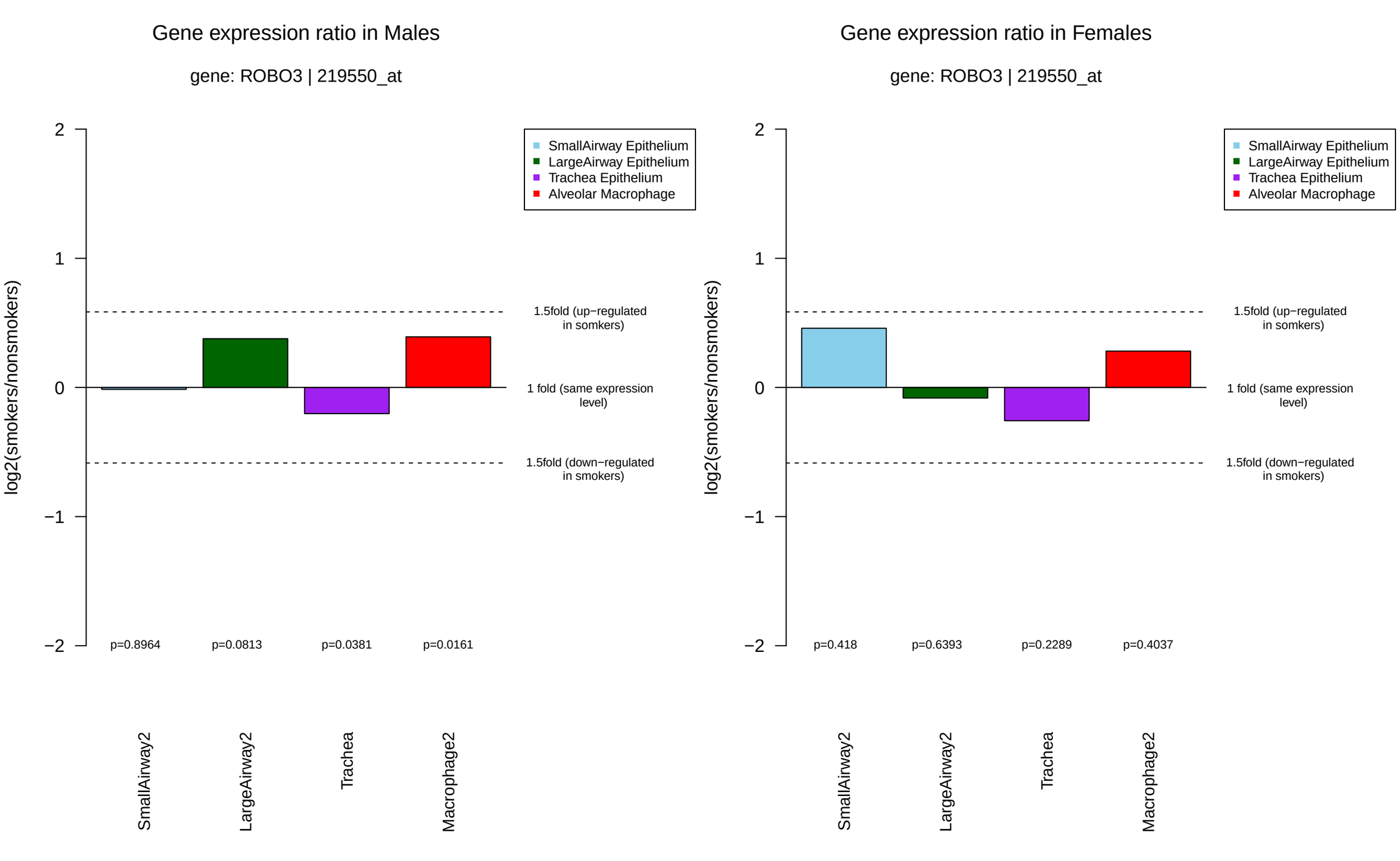
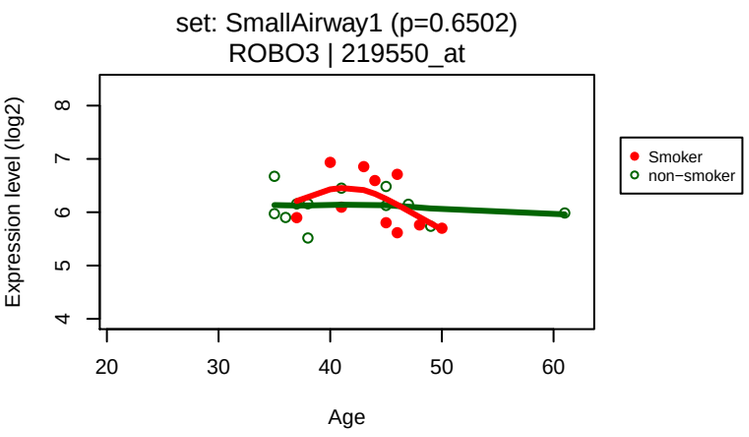
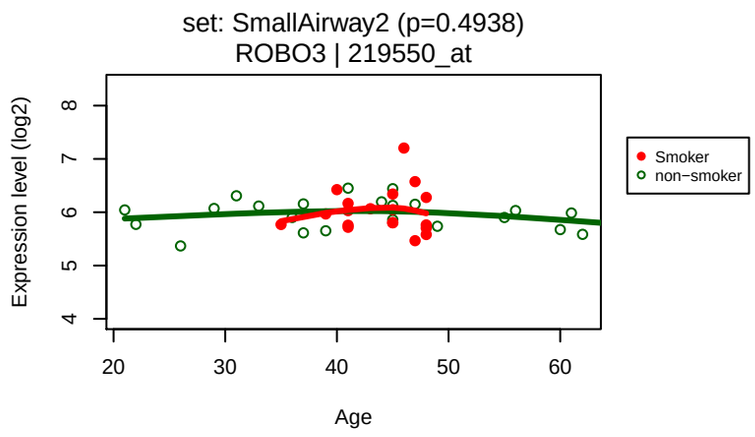
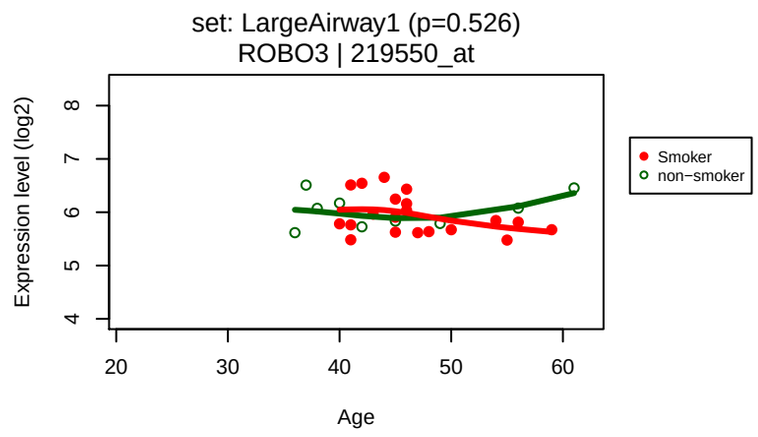

Gene expression of smokers and nonsmokers
gene: ROBO3 | 219550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6502
p=0.4938
p=0.947
p=0.526
p=0.162
p=0.0086
p=0.5915
p=0.0133
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

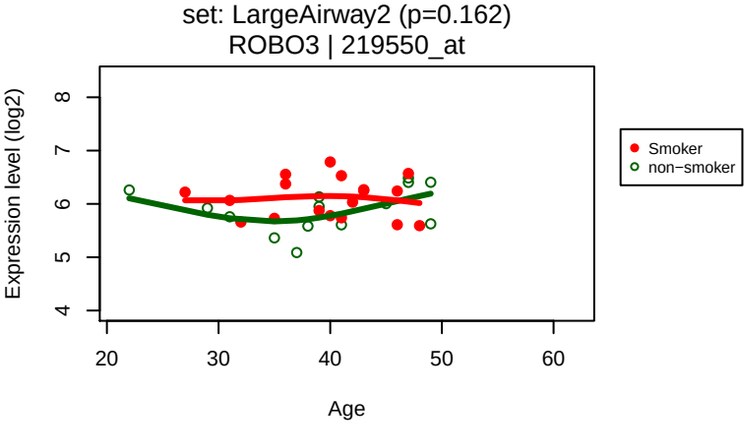

Gene expression ratio (smokers/nonsmokers)
gene: ROBO3 | 219550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6502
p=0.4938
p=0.947
p=0.526
p=0.162
p=0.0086
p=0.5915
p=0.0133
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

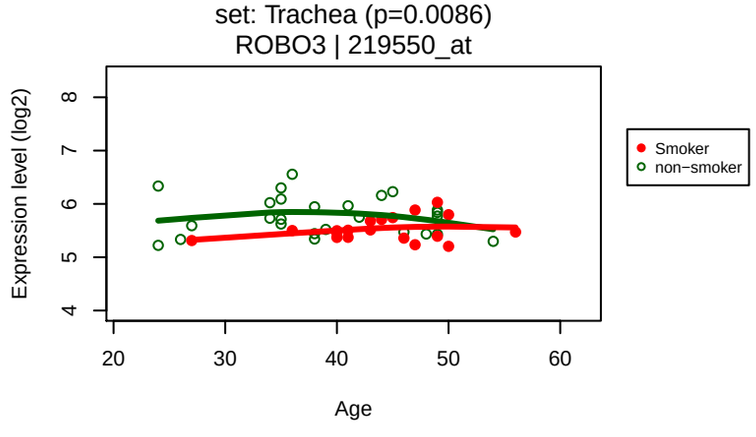

Gene expression ratio in Males
gene: ROBO3 | 219550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8964
p=0.0813
p=0.0381
p=0.0161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROBO3 | 219550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.418
p=0.6393
p=0.2289
p=0.4037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

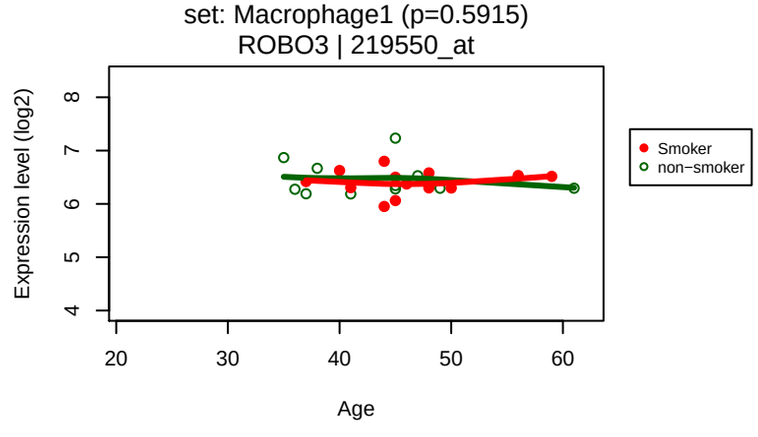

set: SmallAirway1 (p=0.6502)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker

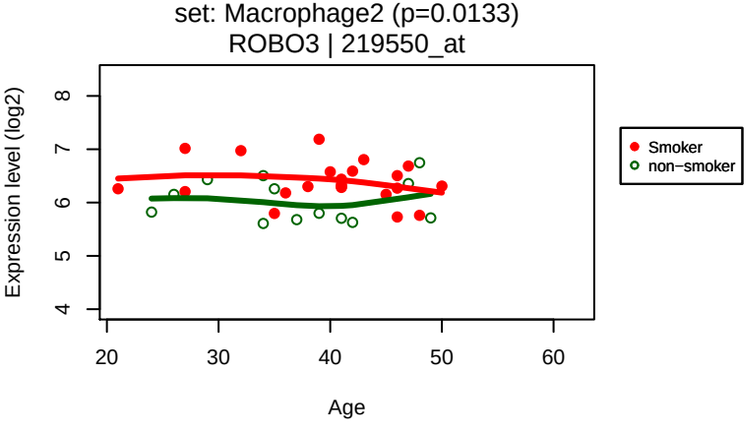

set: SmallAirway2 (p=0.4938)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker

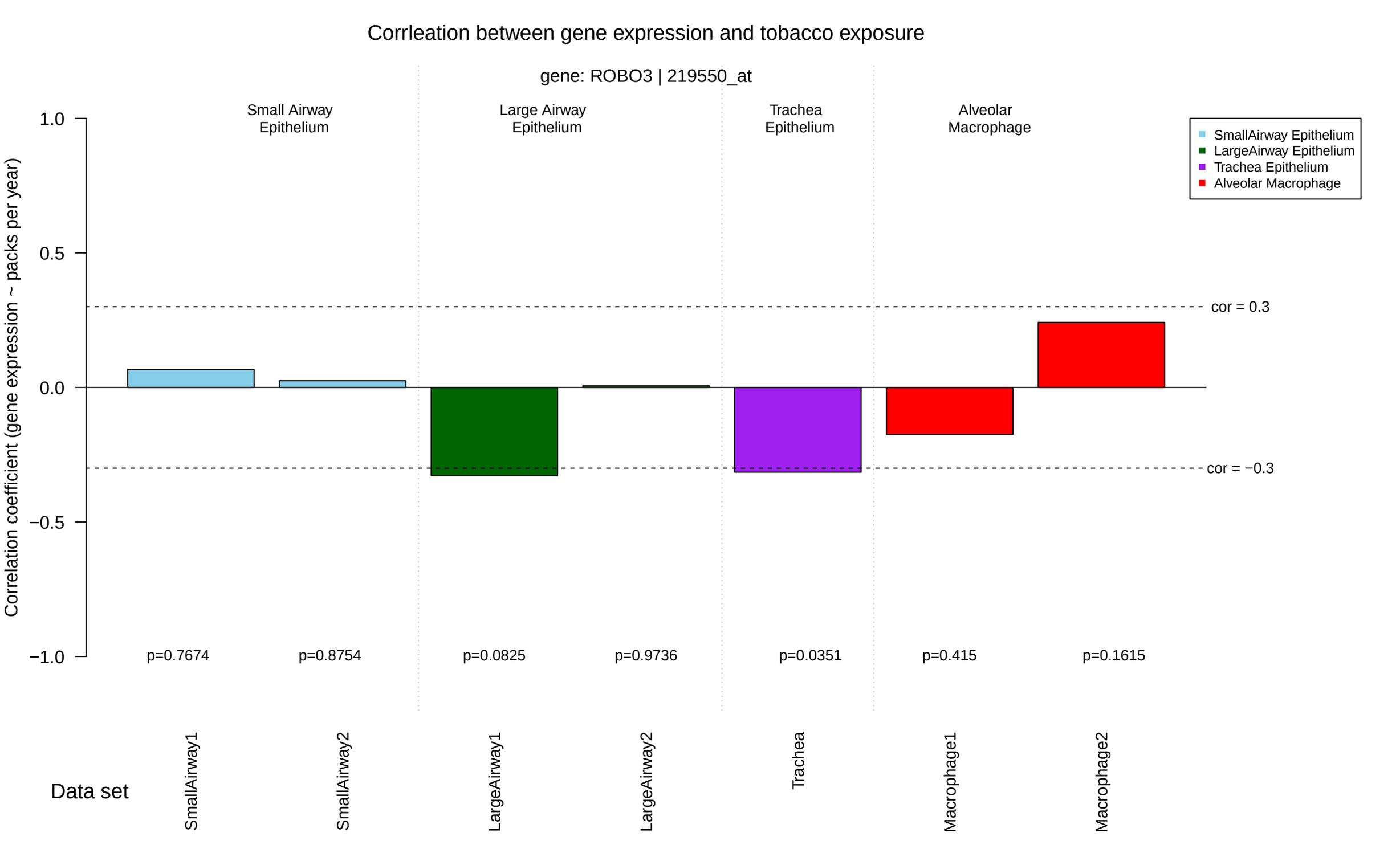

set: LargeAirway1 (p=0.526)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.162)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0086)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5915)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0133)
ROBO3 | 219550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROBO3 | 219550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7674
p=0.8754
p=0.0825
p=0.9736
p=0.0351
p=0.415
p=0.1615
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2